

Figure 5: Algorithm of Annealing

Input: $P_{original}$ Output: $S_{optimal}$

Function SolveQUBOviaQuantumAnnealing($P_{original}$):

// 1: (Problem Transformation)

```
C ← InitializeEmptyQUBOMatrix
Q ← InitializeEmptyQUBOMatrix
for each variable  $v$  in  $P_{original}$  do
    MapDecisionVariable( $v$ ,  $C$ , {0,1})
end
// QUBO Q
ConvertObjectiveFunction( $P_{original}$ ) to  $\sum_{i \leq j} Q_{ij}x_i x_j$ 
// QUBO Q
```

```
for each constraint  $c$  in  $P_{original}$  do
// 2: (Physical Solution)
    PenaltyTerm $_c$  ← FormulateQuadraticPenaltyForc
     $H_{problem}$  ← ConstructQUBOHamiltonianFromQ //  $H_{problem}$ 
    AddPenaltyTerm $_c$  to  $\sum_{i \leq j} Q_{ij}x_i x_j$ 
end
 $H_{initial}$  ←
```

 DefineInitialSimpleHamiltonian(e.g., transverse-field) //

$H_{initial}(t) = (1 - t/T)H_{initial} + (t/T)H_{problem}$ //

 InitializeQuantumSystemInGroundStateOf $H_{initial}$ //

for $t=0$ to T , Δt do

 Apply $H_{total}(t)$ to quantum system

 Allow system to evolve adiabatically

end

// 3:

Measurement ← MeasureQuantumSystemInComputationalBasis

S_{binary} ←

 ExtractBinaryValues $\{x_1, x_2, \dots, x_N\}$ fromMeasurement //

$S_{optimal}$ ← Map S_{binary} backTo $P_{original}$ variables

return $S_{optimal}$